

Interface Capability Brief

EDI 278 Prior Auth · Stakeholder overview · 2026-08-13

1. Executive summary

Ingest X12 278 prior-auth requests, apply completeness rules (member, procedure, diagnosis, attachment when required), simulate a payer decision, and emit an X12 278 response plus an EHR HL7 ORU “auth decided” notice. Demo only — not Da Vinci PAS, not real payer APIs.

2. Who this is for

- Sources: Provider PA drop folder (input/inbound/.edi) - Destinations: decision buckets (approved / denied / incomplete / pended), output/responses/.edi (278), output/ehr-notices/*.hl7 (ORU), SQL AuthRequest status - Demo vs production: Demo only — synthetic 278, catalog rules in SQL, no FHIR PAS, no LLP to a live EHR

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3. What the interface can do

Capabilities below are drawn from the working design and, when present, the published CapabilityStatement — not from aspirational roadmap items.

- Transport: Directory / File (DirectoryListener)
- Format: X12 278 (5010-shaped), multi-ST / multi-Transaction
- Identity: TRN02 → AuthTraceNumber
- Samples: samples/sample_278_prior_auths.edi

Integration highlights

FQCN sources: edi-835-payment-integrity (directory, EDI fork, JDBC, XPath router), medical-device-hl7-ehr (HL7 XML→ER7). Image: pilotfish-eip:23R1.

Stage	Module / mechanism	Notes
R1 Listener	DirectoryListener	Poll \$\$EDI_INBOUND_DIRECTORY
R1 Format	EDITransformationProcessor EDI→XML + XPathForkingModule //Transaction	TableData 278-A1 (5010). XML Formatting before XML file writes.
R1 Target	XPath extract → DatabaseSqlProcessor AuthCatalog lookup → XSLT completeness	Build AuthDecision. SQLXML tags are UPPERCASE.
R1 Transport	DirectoryTransport	Stage under output/staged-decisions/
R2 Listener	DirectoryListener	Poll staged decisions
R2 Router	XPathRoutingModule accumulate=true	Emit 278+ORU+SQL for all; bucket writers by DecisionBucket
R2 278 emit	XSLT → EDITransformationProcessor XML→EDI	XSLT builds XCSDData only; no hardcoded ISA*

Stage	Module / mechanism	Notes
R2 ORU emit	XSLT → HL7TransformationProcessor XML→HL7 2.X	XSLT builds ORU XML only; no hardcoded MSH\
R2 Transports	Directory + Null + SQL UPDATE	Unique processor names per route

4. Honest boundaries (not claimed yet)

- FHIR Da Vinci PAS inbound later? LLP for ORU later?

5. What you can verify in a walkthrough

Use these as checklist items when reviewing the demo with business stakeholders. They describe outcomes, not module names.

- Exercise route “1 - Poll And Score Auths” from the Web UI or documented smoke path
- Confirm outputs land in the documented folders / queues and appear in the Web UI status views

6. How it works (plain language)

7. Validation

- Checked: EDI parses; TRN identity; completeness; catalog disposition - Not checked: full 278 SNIP, Da Vinci PAS, real attachments, LLP - Sample outcomes: PACOMPLETE01 approved, PAINCOMPLETE01 incomplete, PADENY01 denied, PAPEND01 pending

6. State & idempotency

- Status: AuthRequest OPEN → APPROVED \| DENIED \| INCOMPLETE \| PENDED - Dedup: AuthTraceNumber - Unparseable EDI fails closed; incompleteness is business theater (still emits 278/ORU)

8. Dual-write / side effects

- Order: stage AuthDecision → fan-out (278 + ORU + bucket + SQL) - Compensation: none (demo)

3. Inbound contract

- Transport: Directory / File (DirectoryListener) - Format: X12 278 (5010-shaped), multi-ST / multi-Transaction - Identity: TRN02 → AuthTraceNumber - Samples: samples/sample_278_prior_auths.edi

Routes in this interface

Each route is a configured PilotFish flow. Technical wiring appears in the Route Diagrams PDF; here is the stakeholder reading.

1 - Poll And Score Auths: runs steps such as Poll EDI 278 Files, Fork - XPath, Stage Auth Decision.

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2 - Route Auth Decisions: runs steps such as Poll Staged Auth Decisions, Conditional Router, Write 278 Response, Write ORU Notice.

2 - Route Auth Decisions: runs steps such as Poll Staged Auth Decisions, Extract Decision Attributes, Conditional Router, Write 278 Response.

2 - Route Auth Decisions: runs steps such as Poll Staged Auth Decisions, Extract Decision Attributes, Conditional Router, Map Auth Decision To 278 XCSDData.

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7. Security & trust posture

- X12 SNIP validation is used where configured to reject bad EDI before downstream work
- Credentials and passwords in this package are demo-only — not production secrets
- Risk noted in design: High — No real payer / FHIR PAS — Accepted theater
- Risk noted in design: Med — 23R1 trial tables — TableData mount (5 ../)
- Risk noted in design: Med — Duplicate processor names — Unique names per route
- Risk noted in design: Med — SQLXML uppercase tags — XPath matches both cases

8. Seeing it work

- - Ports: SQL 14340, EIP 8120, Web UI 8121 - DB: Edi278PriorAuth, sa / PilotFish_Demo1! - TableData: ../.../EDI/TableData/x12 - Compose project: edi-278-prior-auth
- docker compose --profile full up -d --build
- Web UI (open in Cursor, not Chrome): `http://127.0.0.1:8121/`
- EIP: `http://127.0.0.1:8120/eip/`
- SQL: 127.0.0.1:14340 (sa / PilotFish_Demo1! · Edi278PriorAuth)

9. Related technical documents

- DESIGN.md — working engineering specification for this interface
- README.md — how to run, ports, and smoke commands
- This Capability Brief — shareable stakeholder summary (regenerate after design changes)

This document is generated from the interface sources listed above. If design and runtime diverge, believe the smoke-tested runtime and update DESIGN.md.